LIMIT PRESERVATION OF THE FORMAL-CONTRACTION GLUING FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. The complete530-line author proof is retained; the task concerns the exact contraction functor in Situationcontractions, not all formal completion functors.

Editorial note. The separately prepared0GI9 establishes one smooth neighbourhood from versality; it neither states nor supplies the present descent of completion-valued data.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Noetherian formal algebraic spaces, formal modifications, inverse limits of finitely presented morphisms, proper coherent cohomology and formal functions, Artin–Rees and adic completion. The precise contraction functor and its local solution/equality lemmas are retained.

Source reference. Original proposition 0GI3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20603–20603. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0GE1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20461–20461. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 05N0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7289–7289. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0GE3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20463–20463. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 05ER: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7000–7000. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:artin.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 08AS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10542–10542. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 073K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9064–9064. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 08AZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10549–10549. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0GH5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20570–20570. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0GH4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20569–20569. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0GH6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20571–20571. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AQT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13521–13521. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0GD6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20431–20431. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 01YV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2570–2570. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0CUF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16100–16100. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0A2P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12712–12712. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 01ZC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2587–2587. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0GHZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20599–20599. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 049N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5586–5586. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01Z4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2579–2579. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 0ARB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13539–13539. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0GI4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20604–20604. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0ARW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13559–13559. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 03XX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5164–5164. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01ZM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2599–2599. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 081F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10215–10215. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 081E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10214–10214. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01Z0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2575–2575. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00L6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 776–776. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0BNH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14665–14665. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0GC6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20396–20396. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 0GDK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20445–20445. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0AIX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13280–13280. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 0GE7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20467–20467. Stable Stacks tag for an

expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0GDN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20448–20448. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 0GDU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20454–20454. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *In Situation 0.2 the functor F is limit preserving: for any directed limit $V = \lim V_\lambda$ of Noetherian affine schemes over S we have $F(V) = \operatorname{colim} F(V_\lambda)$.*

Situation 0.2. Let S be a locally Noetherian scheme. Let X' be an algebraic space locally of finite type over S. Let $T' \subset |X'|$ be a closed subset. Let $U' \subset X'$ be the open subspace with $|U'| = |X'| \setminus T'$. Let W be a locally Noetherian formal algebraic space over S with W_{red} locally of finite type over S. Finally, we let

$$g : X'_{/T'} \longrightarrow W$$

be a formal modification, see Algebraization of Formal Spaces, Definition 0GDK. Recall that $X'_{/T'}$ denotes the formal completion of X' along T', see Formal Spaces, Section 0AIX.

In the situation above our goal is to prove that there exists a proper morphism $f : X' \to X$ of algebraic spaces over S, a closed subset $T \subset |X|$, and an isomorphism $a : X_{/T} \to W$ of formal algebraic spaces such that

(1) T' is the inverse image of T by $|f| : |X'| \to |X|$,
(2) $f : X' \to X$ maps U' isomorphically to an open subspace U of X, and
(3) $g = a \circ f_{/T}$ where $f_{/T} : X'_{/T'} \to X_{/T}$ is the induced morphism.

Let us say that $(f : X' \to X, T, a)$ is a *solution*.

We will follow Artin's strategy by constructing a functor F on the category of locally Noetherian schemes over S, showing that F is an algebraic space using Proposition 0GE7, and proving that setting $X = F$ works.

Remark 0.3. In particular, we cannot prove that the desired result is true for every Situation 0.2 because we will need to assume the local rings of S are G-rings. If you can prove the result in general or if you have a counter example, please let us know at stacks.project@gmail.com.

In Situation 0.2 let V be a locally Noetherian scheme over S. The value of our functor F on V will be all triples

$$(Z, u' : V \setminus Z \to U', \hat{x} : V_{/Z} \to W)$$

satisfying the following conditions

(1) $Z \subset V$ is a closed subset,
(2) $u' : V \setminus Z \to U'$ is a morphism over S,
(3) $\hat{x} : V_{/Z} \to W$ is an adic morphism of formal algebraic spaces over S,
(4) u' and $\hat{x}$ are compatible (see below).

The compatibility condition is the following: pulling back the formal modification g we obtain a formal modification

$$X'_{/T'} \times_{g,W,\hat{x}} V_{/Z} \longrightarrow V_{/Z}$$

See Algebraization of Formal Spaces, Lemma 0GDN. By the main theorem on dilatations (Algebraization of Formal Spaces, Theorem 0GDU), there is a unique proper morphism $V' \to V$ of algebraic spaces which is an isomorphism over $V \setminus Z$ such that $V'_{/Z} \to V_{/Z}$ is isomorphic to the displayed arrow. In other words, for some morphism $\hat{x}' : V'_{/Z} \to X'_{/T'}$ we have a cartesian diagram

$$\begin{array}{ccc} V'_{/Z} & \longrightarrow & V_{/Z} \\ {\scriptstyle \hat{x}'}\downarrow & & \downarrow{\scriptstyle \hat{x}} \\ X'_{/T'} & \xrightarrow{g} & W \end{array}$$

of formal algebraic spaces. We will think of $V \setminus Z$ as an open subspace of V' without further mention. The compatibility condition is that there should be a morphism $x' : V' \to X'$ restricting to u' and $\hat{x}$ over $V \setminus Z \subset V'$ and $V'_{/Z}$. In other words, such that the diagram

$$\begin{array}{ccccccc} V \setminus Z & \longrightarrow & V' & \longleftarrow & V'_{/Z} & \longrightarrow & V_{/Z} \\ {\scriptstyle u'}\downarrow & & \downarrow{\scriptstyle x'} & & \downarrow{\scriptstyle \hat{x}'} & & \downarrow{\scriptstyle \hat{x}} \\ U' & \longrightarrow & X' & \longleftarrow & X'_{/T'} & \xrightarrow{g} & W \end{array}$$

is commutative. Observe that by Algebraization of Formal Spaces,

if it exists. We will indicate this situation by saying "$V' \to V$, $\hat{x}'$, *and* x' *witness the compatibility between* u' *and* $\hat{x}$".

Remark 0.4. In Situation 0.2 let V be a locally Noetherian scheme over S. Let $(Z, u', \hat{x})$ be a triple satisfying (1), (2), and (3) above. We want to explain a way to think about the compatibility condition (4). It will not be mathematically precise as we are going use a fictitious category An_S of analytic spaces over S and a fictitious analytification functor

$$\left\{\begin{array}{l}\text{locally Noetherian formal} \\ \text{algebraic spaces over } S\end{array}\right\} \longrightarrow An_S, \qquad Y \longmapsto Y^{an}$$

For example if $Y = \mathrm{Spf}(k[[t]])$ over $S = \mathrm{Spec}(k)$, then Y^{an} should be thought of as an open unit disc. If $Y = \mathrm{Spec}(k)$, then Y^{an} is a single point. The category An_S should have open and closed immersions and we should be able to take the open complement of a closed. Given Y the morphism $Y_{red} \to Y$ should induces a closed immersion $Y^{an}_{red} \to Y^{an}$. We set $Y^{rig} = Y^{an} \setminus Y^{an}_{red}$ equal to its open complement. If Y is an algebraic space and if $Z \subset Y$ is closed, then the morphism $Y_{/Z} \to Y$ should induce an open immersion $Y^{an}_{/Z} \to Y^{an}$ which in turn should induce an open immersion

$$can : (Y_{/Z})^{rig} \longrightarrow (Y \setminus Z)^{an}$$

Also, given a formal modification $g : Y' \to Y$ of locally Noetherian formal algebraic spaces, the induced morphism $g^{rig} : (Y')^{rig} \to Y^{rig}$ should be an isomorphism. Given An_S and the analytification functor, we can consider the requirement that

$$\begin{array}{ccc} (V_{/Z})^{rig} & \xrightarrow[can]{} & (V \setminus Z)^{an} \\ {\scriptstyle (g^{rig})^{-1} \circ \hat{x}^{an}}\downarrow & & \downarrow{\scriptstyle (u')^{an}} \\ (X'_{/T'})^{rig} & \xrightarrow{can} & (X' \setminus T')^{an} \end{array}$$

commutes. This makes sense as $g^{rig} : (X'_{T'})^{rig} \to W^{rig}$ is an isomorphism and $U' = X' \setminus T'$. Finally, under some assumptions of faithfulness of the analytification functor, this requirement will be equivalent to the compatibility condition formulated above. We hope this will motivate the reader to think of the compatibility of u' and $\hat{x}$ as the requirement that some maps be equal, rather than asking for the existence of a certain commutative diagram.

Lemma 0.5. *In Situation 0.2 the rule F that sends a locally Noetherian scheme V over S to the set of triples $(Z, u', \hat{x})$ satisfying the compatibility condition and which sends a a morphism $\varphi : V_2 \to V_1$ of locally Noetherian schemes over S to the map*

$$F(\varphi) : F(V_1) \longrightarrow F(V_2)$$

sending an element $(Z_1, u'_1, \hat{x}_1)$ of $F(V_1)$ to $(Z_2, u'_2, \hat{x}_2)$ in $F(V_2)$ given by

1. *$Z_2 \subset V_2$ is the inverse image of Z_1 by φ,*
2. *u'_2 is the composition of u'_1 and $\varphi|_{V_2 \setminus Z_2} : V_2 \setminus Z_2 \to V_1 \setminus Z_1$,*
3. *$\hat{x}_2$ is the composition of $\hat{x}_1$ and $\varphi_{/Z_2} : V_{2,/Z_2} \to V_{1,/Z_1}$*

is a contravariant functor.

Proof. To see the compatibility condition between u'_2 and $\hat{x}_2$, let $V'_1 \to V_1$, $\hat{x}'_1$, and x'_1 witness the compatibility between u'_1 and $\hat{x}_1$. Set $V'_2 = V_2 \times_{V_1} V'_1$, set $\hat{x}'_2$ equal to the composition of $\hat{x}'_1$ and $V'_{2,/Z_2} \to V'_{1,/Z_1}$, and set x'_2 equal to the composition of x'_1 and $V'_2 \to V'_1$. Then $V'_2 \to V_2$, $\hat{x}'_2$, and x'_2 witness the compatibility between u'_2 and $\hat{x}_2$. We omit the detailed verification. □

Lemma 0.6. *In Situation 0.2 if there exists a solution $(f : X' \to X, T, a)$ then there is a functorial bijection $F(V) = \mathrm{Mor}_S(V, X)$ on the category of locally Noetherian schemes V over S.*

Proof. Let V be a locally Noetherian scheme over S. Let $x : V \to X$ be a morphism over S. Then we get an element $(Z, u', \hat{x})$ in $F(V)$ as follows

1. $Z \subset V$ is the inverse image of T by x,
2. $u' : V \setminus Z \to U' = U$ is the restriction of x to $V \setminus Z$,
3. $\hat{x} : V_{/Z} \to W$ is the composition of $x_{/Z} : V_{/Z} \to X_{/T}$ with the isomorphism $a : X_{/T} \to W$.

This triple satisfies the compatibility condition because we can take $V' = V \times_{x,X} X'$, we can take $\hat{x}'$ the completion of the projection $x' : V' \to X'$.

Conversely, suppose given an element $(Z, u', \hat{x})$ of $F(V)$. We claim there is a unique morphism $x : V \to X$ compatible with u' and $\hat{x}$. Namely, let $V' \to V$, $\hat{x}'$, and x' witness the compatibility between u' and $\hat{x}$. Then Algebraization of Formal Spaces, Proposition 0GI3 is exactly the result we need to find a unique morphism $x : V \to X$ agreeing with $\hat{x}$ over $V_{/Z}$ and with x' over V' (and a fortiori agreeing with u' over $V \setminus Z$).

We omit the verification that the two constructions above define inverse bijections between their respective domains. □

Lemma 0.7. *In Situation 0.2 if there exists an algebraic space X locally of finite type over S and a functorial bijection $F(V) = \mathrm{Mor}_S(V, X)$ on the category of locally Noetherian schemes V over S, then X is a solution.*

Proof. We have to construct a proper morphism $f : X' \to X$, a closed subset $T \subset |X|$, and an isomorphism $a : X_{/T} \to W$ with properties (1), (2), (3) listed just below Situation 0.2.

The discussion in this proof is a bit pedantic because we want to carefully match the underlying categories. In this paragraph we explain how the adventurous reader

can proceed less timidly. Namely, the reader may extend our definition of the functor F to all locally Noetherian algebraic spaces over S. Doing so the reader may then conclude that F and X agree as functors on the category of these algebraic spaces, i.e., X represents F. Then one considers the universal object $(T, u', \hat{x})$ in $F(X)$. Then the reader will find that for the triple $X'' \to X$, $\hat{x}'$, x' witnessing the compatibility between u' and $\hat{x}$ the morphism $x' : X'' \to X'$ is an isomorphism and this will produce $f : X' \to X$ by inverting x'. Finally, we already have $T \subset |X|$ and the reader may show that $\hat{x}$ is an isomorphism which can served as the last ingredient namely a.

Denote $h_X(-) = \mathrm{Mor}_S(-, X)$ the functor of points of X restricted to the category $(\mathit{Noetherian}/S)_{\mathit{étale}}$ of Section 0GE1. By Limits of Spaces, Remark 05N0 the algebraic spaces X and X' are limit preserving. Hence so are the restrictions h_X and $h_{X'}$. To construct f it therefore suffices to construct a transformation $h_{X'} \to h_X = F$, see Lemma 0GE3. Thus let $V \to S$ be an object of $(\mathit{Noetherian}/S)_{\mathit{étale}}$ and let $\tilde{x} : V \to X'$ be in $h_{X'}(V)$. Then we get an element $(Z, u', \hat{x})$ in $F(V)$ as follows

(1) $Z \subset V$ is the inverse image of T' by $\tilde{x}$,
(2) $u' : V \setminus Z \to U'$ is the restriction of $\tilde{x}$ to $V \setminus Z$,
(3) $\hat{x} : V_{/Z} \to W$ is the composition of $x_{/Z} : V_{/Z} \to X'_{/T'}$ with $g : X'_{/T'} \to W$.

This triple satisfies the compatibility condition: first we always obtain $V' \to V$ and $\hat{x}' : V'_{/Z'} \to X'_{/T'}$ for free (see discussion preceding Lemma 0.5). Then we just define $x' : V' \to X'$ to be the composition of $V' \to V$ and the morphism $\tilde{x} : V \to X'$. We omit the verification that this works.

If $\xi : V \to X$ is an étale morphism where V is a scheme, then we obtain $\xi = (Z, u', \hat{x}) \in F(V) = h_X(V) = X(V)$. Of course, if $\varphi : V' \to V$ is a further étale morphism of schemes, then $(Z, u', \hat{x})$ pulled back to $F(V')$ corresponds to $\xi \circ \varphi$. The closed subset $T \subset |X|$ is just defined as the closed subset such that $\xi : V \to X$ for $\xi = (Z, u', \hat{x})$ pulls T back to Z

Consider Noetherian schemes V over S and a morphism $\xi : V \to X$ corresponding to $(Z, u', \hat{x})$ as above. Then we see that $\xi(V)$ is set theoretically contained in T if and only if $V = Z$ (as topological spaces). Hence we see that $X_{/T}$ agrees with W as a functor. This produces the isomorphism $a : X_{/T} \to W$. (We've omitted a small detail here which is that for the locally Noetherian formal algebraic spaces $X_{/T}$ and W it suffices to check one gets an isomorphism after evaluating on locally Noetherian schemes over S.)

We omit the proof of conditions (1), (2), and (3). □

Remark 0.8. In Situation 0.2. Let V be a locally Noetherian scheme over S. Let $(Z_i, u'_i, \hat{x}_i) \in F(V)$ for $i = 1, 2$. Let $V'_i \to V$, $\hat{x}'_i$ and x'_i witness the compatibility between u'_i and $\hat{x}_i$ for $i = 1, 2$.

Set $V' = V'_1 \times_V V'_2$. Let $E' \to V'$ denote the equalizer of the morphisms

$$V' \to V'_1 \xrightarrow{x'_1} X' \quad \text{and} \quad V' \to V'_2 \xrightarrow{x'_2} X'$$

Set $Z = Z_1 \cap Z_2$. Let $E_W \to V_{/Z}$ be the equalizer of the morphisms

$$V_{/Z} \to V_{/Z_1} \xrightarrow{\hat{x}_1} W \quad \text{and} \quad V_{/Z} \to V_{/Z_2} \xrightarrow{\hat{x}_2} W$$

Observe that $E' \to V$ is separated and locally of finite type and that E_W is a locally Noetherian formal algebraic space separated over V. The compatibilities between the various morphisms involved show that

(1) $\mathrm{Im}(E' \to V) \cap (Z_1 \cup Z_2)$ is contained in $Z = Z_1 \cap Z_2$,

(2) the morphism $E' \times_V (V \setminus Z) \to V \setminus Z$ is a monomorphism and is equal to the equalizer of the restrictions of u'_1 and u'_2 to $V \setminus (Z_1 \cup Z_2)$,
(3) the morphism $E'_{/Z} \to V_{/Z}$ factors through E_W and the diagram

$$\begin{array}{ccc} E'_{/Z} & \longrightarrow & X'_{/T'} \\ \downarrow & & \downarrow g \\ E_W & \longrightarrow & W \end{array}$$

is cartesian. In particular, the morphism $E'_{/Z} \to E_W$ is a formal modification as the base change of g,
(4) E', $(E' \to V)^{-1}Z$, and $E'_{/Z} \to E_W$ is a triple as in Situation 0.2 with base scheme the locally Noetherian scheme V,
(5) given a morphism $\varphi : A \to V$ of locally Noetherian schemes, the following are equivalent
   (a) $(Z_1, u'_1, \hat{x}_1)$ and $(Z_2, u'_2, \hat{x}_2)$ restrict to the same element of $F(A)$,
   (b) $A \setminus \varphi^{-1}(Z) \to V \setminus Z$ factors through $E' \times_V (V \setminus Z)$ and $A_{/\varphi^{-1}(Z)} \to V_{/Z}$ factors through E_W.

We conclude, using Lemmas 0.6 and 0.7, that if there is a solution $E \to V$ for the triple in (4), then E represents $F \times_{\Delta, F \times F} V$ on the category of locally Noetherian schemes over V.

Lemma 0.9. *In Situation 0.2 assume given a closed subset $Z \subset S$ such that*

(1) *the inverse image of Z in X' is T',*
(2) *$U' \to S \setminus Z$ is a closed immersion,*
(3) *$W \to S_{/Z}$ is a closed immersion.*

Then there exists a solution $(f : X' \to X, T, a)$ and moreover $X \to S$ is a closed immersion.

Proof. Suppose we have a closed subscheme $X \subset S$ such that $X \cap (S \setminus Z) = U'$ and $X_{/Z} = W$. Then X represents the functor F (some details omitted) and hence is a solution. To find X is clearly a local question on S. In this way we reduce to the case discussed in the next paragraph.

Assume $S = \mathrm{Spec}(A)$ is affine. Let $I \subset A$ be the radical ideal cutting out Z. Write $I = (f_1, \ldots, f_r)$. By assumption we are given

(1) the closed immersion $U' \to S \setminus Z$ determines ideals $J_i \subset A[1/f_i]$ such that J_i and J_j generate the same ideal in $A[1/f_i f_j]$,
(2) the closed immersion $W \to S_{/Z}$ is the map $\mathrm{Spf}(A^\wedge/J') \to \mathrm{Spf}(A^\wedge)$ for some ideal $J' \subset A^\wedge$ in the I-adic completion $A^\wedge$ of A.

To finish the proof we need to find an ideal $J \subset A$ such that $J_i = J[1/f_i]$ and $J' = JA^\wedge$. By More on Algebra, Proposition 05ER it suffices to show that J_i and J' generate the same ideal in $A^\wedge[1/f_i]$ for all i.

Recall that $A' = H^0(X', \mathcal{O})$ is a finite A-algebra whose formation commutes with flat base change (Cohomology of Spaces, Lemmas 08AS and 073K). Denote $J'' = \mathrm{Ker}(A \to A')$[1]. We have $J_i = J''A[1/f_i]$ as follows from base change to the spectrum of $A[1/f_i]$. Observe that we have a commutative diagram

$$\begin{array}{ccccc} X' & \longleftarrow & X'_{/T'} \times_{S_{/Z}} \mathrm{Spf}(A^\wedge) & = & X'_{/T'} \times_W \mathrm{Spf}(A^\wedge/J') \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \longleftarrow & \mathrm{Spf}(A^\wedge) & \longleftarrow & \mathrm{Spf}(A^\wedge/J') \end{array}$$

[1]Contrary to what the reader may expect, the ideals J and J'' won't agree in general.

The middle vertical arrow is the completion of the left vertical arrow along the obvious closed subsets. By the theorem on formal functions we have

$$(A')^\wedge = \Gamma(X' \times_S \mathrm{Spec}(A^\wedge), \mathcal{O}) = \lim H^0(X' \times_S \mathrm{Spec}(A/I^n), \mathcal{O})$$

See Cohomology of Spaces, Theorem 08AZ. From the diagram we conclude that J' maps to zero in $(A')^\wedge$. Hence $J' \subset J''A^\wedge$. Consider the arrows

$$X'_{/T'} \to \mathrm{Spf}(A^\wedge/J''A^\wedge) \to \mathrm{Spf}(A^\wedge/J') = W$$

We know the composition g is a formal modification (in particular rig-étale and rig-surjective) and the second arrow is a closed immersion (in particular an adic monomorphism). Hence $X'_{/T'} \to \mathrm{Spf}(A^\wedge/J''A^\wedge)$ is rig-surjective and rig-étale, see Algebraization of Formal Spaces, Lemmas 0GH5 and 0GH4. Applying Algebraization of Formal Spaces, Lemmas 0GH6 and 0AQT we conclude that $\mathrm{Spf}(A^\wedge/J''A^\wedge) \to W$ is rig-étale and rig-surjective. By Algebraization of Formal Spaces, Lemma 0GD6 we conclude that $I^nJ''A^\wedge \subset J'$ for some $n > 0$. It follows that $J''A^\wedge[1/f_i] = J'A^\wedge[1/f_i]$ and we deduce $J_iA^\wedge[1/f_i] = J'A^\wedge[1/f_i]$ for all i as desired. □

Lemma 0.10. *In Situation 0.2 assume $X' \to S$ and $W \to S$ are separated. Then the diagonal $\Delta : F \to F \times F$ is representable by closed immersions.*

Proof. Combine Lemma 0.9 with the discussion in Remark 0.8. □

Lemma 0.11. *In Situation 0.2 the functor F satisfies the sheaf property for all étale coverings of locally Noetherian schemes over S.*

Proof. Omitted. Hint: morphisms may be defined étale locally. □

Proof. This is an absurdly long proof. Much of it consists of standard arguments on limits and étale localization. We urge the reader to skip ahead to the last part of the proof where something interesting happens.

Let $V = \lim_{\lambda \in \Lambda} V_i$ be a directed limit of schemes over S with V and V_λ Noetherian and with affine transition morphisms. See Limits, Section 01YV for material on limits of schemes. We will prove that $\mathrm{colim}\, F(V_\lambda) \to F(V)$ is bijective.

Proof of injectivity: notation. Let $\lambda \in \Lambda$ and $\xi_{\lambda,1}, \xi_{\lambda,2} \in F(V_\lambda)$ be elements which restrict to the same element of $F(V)$. Write $\xi_{\lambda,1} = (Z_{\lambda,1}, u'_{\lambda,1}, \hat{x}_{\lambda,1})$ and $\xi_{\lambda,2} = (Z_{\lambda,2}, u'_{\lambda,2}, \hat{x}_{\lambda,2})$.

Proof of injectivity: agreement of $Z_{\lambda,i}$. Since $Z_{\lambda,1}$ and $Z_{\lambda,2}$ restrict to the same closed subset of V, we may after increasing i assume $Z_{\lambda,1} = Z_{\lambda,2}$, see Limits, Lemma 0CUF and Topology, Lemma 0A2P. Let us denote the common value $Z_\lambda \subset V_\lambda$, for $\mu \geq \lambda$ denote $Z_\mu \subset V_\mu$ the inverse image in V_μ and and denote Z the inverse image in V. We will use below that $Z = \lim_{\mu \geq \lambda} Z_\mu$ as schemes if we view Z and Z_μ as reduced closed subschemes.

Proof of injectivity: agreement of $u'_{\lambda,i}$. Since U' is locally of finite type over S and since the restrictions of $u'_{\lambda,1}$ and $u'_{\lambda,2}$ to $V \setminus Z$ are the same, we may after increasing λ assume $u'_{\lambda,1} = u'_{\lambda,2}$, see Limits, Proposition 01ZC. Let us denote the common value u'_λ and denote u' the restriction to $V \setminus Z$.

Proof of injectivity: restatement. At this point we have $\xi_{\lambda,1} = (Z_\lambda, u'_\lambda, \hat{x}_{\lambda,1})$ and $\xi_{\lambda,2} = (Z_\lambda, u'_\lambda, \hat{x}_{\lambda,2})$. The main problem we face in this part of the proof is to show that the morphisms $\hat{x}_{\lambda,1}$ and $\hat{x}_{\lambda,2}$ become the same after increasing λ.

Proof of injectivity: agreement of $\hat{x}_{\lambda,i}|_{Z_\lambda}$. Consider the morpisms $\hat{x}_{\lambda,1}|_{Z_\lambda}, \hat{x}_{\lambda,2}|_{Z_\lambda} : Z_\lambda \to W_{red}$. These morphisms restrict to the same morphism $Z \to W_{red}$. Since W_{red} is a scheme locally of finite type over S we see using Limits, Proposition 01ZC

that after replacing λ by a bigger index we may assume $\hat{x}_{\lambda,1}|_{Z_\lambda} = \hat{x}_{\lambda,2}|_{Z_\lambda} : Z_\lambda \to W_{red}$.

Proof of injectivity: end. Next, we are going to apply the discussion in Remark 0.8 to V_λ and the two elements $\xi_{\lambda,1}, \xi_{\lambda,2} \in F(V_\lambda)$. This gives us

(1) $e_\lambda : E'_\lambda \to V_\lambda$ separated and locally of finite type,
(2) $e_\lambda^{-1}(V_\lambda \setminus Z_\lambda) \to V_\lambda \setminus Z_\lambda$ is an isomorphism,
(3) a monomorphism $E_{W,\lambda} \to V_{\lambda,/Z_\lambda}$ which is the equalizer of $\hat{x}_{\lambda,1}$ and $\hat{x}_{\lambda,2}$,
(4) a formal modification $E'_{\lambda,/Z_\lambda} \to E_{W,\lambda}$

Assertion (2) holds by assertion (2) in Remark 0.8 and the preparatory work we did above getting $u'_{\lambda,1} = u'_{\lambda,2} = u'_\lambda$. Since $Z_\lambda = (V_{\lambda,/Z_\lambda})_{red}$ factors through $E_{W,\lambda}$ because $\hat{x}_{\lambda,1}|_{Z_\lambda} = \hat{x}_{\lambda,2}|_{Z_\lambda}$ we see from Formal Spaces, Lemma 0GHZ that $E_{W,\lambda} \to V_{\lambda,/Z_\lambda}$ is a closed immersion. Then we see from assertion (4) in Remark 0.8 and Lemma 0.9 applied to the triple E'_λ, $e_\lambda^{-1}(Z_\lambda)$, $E'_{\lambda,/Z_\lambda} \to E_{W,\lambda}$ over V_λ that there exists a closed immersion $E_\lambda \to V_\lambda$ which is a solution for this triple. Next we use assertion (5) in Remark 0.8 which combined with Lemma 0.6 says that E_λ is the "equalizer" of $\xi_{\lambda,1}$ and $\xi_{\lambda,2}$. In particular, we see that $V \to V_\lambda$ factors through E_λ. Then using Limits, Proposition 01ZC once more we find $\mu \geq \lambda$ such that $V_\mu \to V_\lambda$ factors through E_λ and hence the pullbacks of $\xi_{\lambda,1}$ and $\xi_{\lambda,2}$ to V_μ are the same as desired.

Proof of surjectivity: statement. Let $\xi = (Z, u', \hat{x})$ be an element of $F(V)$. We have to find a $\lambda \in \Lambda$ and an element $\xi_\lambda \in F(V_\lambda)$ restricting to ξ.

Proof of surjectivity: the question is étale local. By the unicity proved in the previous part of the proof and by the sheaf property of F in Lemma 0.11, the problem is local on V in the étale topology. More precisely, let $v \in V$. We claim it suffices to find an étale morphism $(\tilde{V}, \tilde{v}) \to (V, v)$ and some λ, some an étale morphism $\tilde{V}_\lambda \to V_\lambda$, and some element $\tilde{\xi}_\lambda \in F(\tilde{V}_\lambda)$ such that $\tilde{V} = \tilde{V}_\lambda \times_{V_\lambda} V$ and $\xi|_U = \tilde{\xi}_\lambda|_U$. We omit a detailed proof of this claim[2].

Proof of surjectivity: rephrasing the problem. Recall that any étale morphism $(\tilde{V}, \tilde{v}) \to (V, v)$ with $\tilde{V}$ affine is the base change of an étale morphism $\tilde{V}_\lambda \to V_\lambda$ with $\tilde{V}_\lambda$ affine for some λ, see for example Topologies, Lemma 049N. Given $\tilde{V}_\lambda$ we have $\tilde{V} = \lim_{\mu \geq \lambda} \tilde{V}_\lambda \times_{V_\lambda} V_\mu$. Hence given $(\tilde{V}, \tilde{v}) \to (V, v)$ étale with $\tilde{V}$ affine, we may replace (V, v) by $(\tilde{V}, \tilde{v})$ and ξ by the restriction of ξ to $\tilde{V}$.

Proof of surjectivity: reduce to base being affine. In particular, suppose $\tilde{S} \subset S$ is an affine open subscheme such that $v \in V$ maps to a point of $\tilde{S}$. Then we may according to the previous paragraph, replace V by $\tilde{V} = \tilde{S} \times_S V$. Of course, if we do this, it suffices to solve the problem for the functor F restricted to the category of locally Noetherian schemes over $\tilde{S}$. This functor is of course the functor associated to the whole situation base changed to $\tilde{S}$. Thus we may and do assume $S = \mathrm{Spec}(R)$ is a Noetherian affine scheme for the rest of the proof.

Proof of surjectivity: easy case. If $v \in V \setminus Z$, then we can take $\tilde{V} = V \setminus Z$. This descends to an open subscheme $\tilde{V}_\lambda \subset V_\lambda$ for some λ by Limits, Lemma 01Z4. Next, after increasing λ we may assume there is a morphism $u'_\lambda : \tilde{V}_\lambda \to U'$ restricting to u'. Taking $\tilde{\xi}_\lambda = (\emptyset, u'_\lambda, \emptyset)$ gives the desired element of $F(\tilde{V}_\lambda)$.

[2]To prove this one assembles a collection of the morphisms $\tilde{V} \to V$ into a finite étale covering and shows that the corresponding morphisms $\tilde{V}_\lambda \to V_\lambda$ form an étale covering as well (after increasing λ). Next one uses the injectivity to see that the elements $\tilde{\xi}_\lambda$ glue (after increasing λ) and one uses the sheaf property for F to descend these elements to an element of $F(V_\lambda)$.

Proof of surjectivity: hard case and reduction to affine W. The most difficult case comes from considering $v \in Z \subset V$. We claim that we can reduce this to the case where W is an affine formal scheme; we urge the reader to skip this argument[3]. Namely, we can choose an étale morphism $\tilde{W} \to W$ where $\tilde{W}$ is an affine formal algebraic space such that the image of v by $\hat{x} : V_{/Z} \to W$ is in the image of $\tilde{W} \to W$ (on reductions). Then the morphisms

$$p : \tilde{W} \times_{W,g} X'_{/T'} \longrightarrow X'_{/T'}$$

and

$$q : \tilde{W} \times_{W,\hat{x}} V_{/Z} \to V_{/Z}$$

are étale morphisms of locally Noetherian formal algebraic spaces. By (an easy case of) Algebraization of Formal Spaces, Theorem 0ARB there exists a morphism $\tilde{X}' \to X'$ of algebraic spaces which is locally of finite type, is an isomorphism over U', and such that $\tilde{X}'_{/T'} \to X'_{/T'}$ is isomorphic to p. By Algebraization of Formal Spaces, Lemma 0GI4 the morphism $\tilde{X}' \to X'$ is étale. Denote $\tilde{T}' \subset |\tilde{X}'|$ the inverse image of T'. Denote $\tilde{U}' \subset \tilde{X}'$ the complementary open subspace. Denote $\tilde{g}' : \tilde{X}'_{/\tilde{T}'} \to \tilde{W}$ the formal modification which is the base change of g by $\tilde{W} \to W$. Then we see that

$$\tilde{X}',\ \tilde{T}',\ \tilde{U}',\ \tilde{W},\ \tilde{g} : \tilde{X}'_{/\tilde{T}'} \to \tilde{W}$$

is another example of Situation 0.2. Denote $\tilde{F}$ the functor constructed from this triple. There is a transformation of functors

$$\tilde{F} \longrightarrow F$$

constructed using the morphisms $\tilde{X}' \to X'$ and $\tilde{W} \to W$ in the obvious manner; details omitted.

Proof of surjectivity: hard case and reduction to affine W, part 2. By the same theorem as used above, there exists a morphism $\tilde{V} \to V$ of algebraic spaces which is locally of finite type, is an isomorphism over $V \setminus Z$ and such that $\tilde{V}_{/Z} \to V_{/Z}$ is isomorphic to q. Denote $\tilde{Z} \subset \tilde{V}$ the inverse image of Z. By Algebraization of Formal Spaces, Lemmas 0GI4 and 0ARW the morphism $\tilde{V} \to V$ is étale and separated. In particular $\tilde{V}$ is a (locally Noetherian) scheme, see for example Morphisms of Spaces, Proposition 03XX. We have the morphism u' which we may view as a morphism

$$\tilde{u}' : \tilde{V} \setminus \tilde{Z} \longrightarrow \tilde{U}'$$

where $\tilde{U}' \subset \tilde{X}'$ is the open mapping isomorphically to U'. We have a morphism

$$\tilde{\hat{x}} : \tilde{V}_{/\tilde{Z}} = \tilde{W} \times_{W,\hat{x}} V_{/Z} \longrightarrow \tilde{W}$$

Namely, here we just use the projection. Thus we have the triple

$$\tilde{\xi} = (\tilde{Z}, \tilde{u}', \tilde{\hat{x}}) \in \tilde{F}(\tilde{V})$$

We omit proving the compatibility condition; hints: if $V' \to V$, $\hat{x}'$, and x' witness the compatibility between u' and $\hat{x}$, then one sets $\tilde{V}' = V' \times_V \tilde{V}$ which comes with morphsms $\tilde{\hat{x}}'$ and $\tilde{x}'$ and show this works. The image of $\tilde{\xi}$ under the transformation $\tilde{F} \to F$ is the restriction of ξ to $\tilde{V}$.

Proof of surjectivity: hard case and reduction to affine W, part 3. By our choice of $\tilde{W} \to W$, there is an affine open $\tilde{V}_{open} \subset \tilde{V}$ (we're running out of notation) whose image in V contains our chosen point $v \in V$. Now by the case studied in the next

[3]Artin's approach to the proof of this lemma is to work around this and consequently he can avoid proving the injectivity first. Namely, Artin consistently works with a finite affine étale coverings of all spaces in sight keeping track of the maps between them during the proof. In hindsight that might be preferable to what we do here.

paragraph and the remarks made earlier, we can descend $\tilde{\xi}|_{\tilde{V}_{open}}$ to some element $\tilde{\xi}_\lambda$ of $\tilde{F}$ over $\tilde{V}_{\lambda,open}$ for some étale morphism $\tilde{V}_{\lambda,open} \to V_\lambda$ whose base change to V is $\tilde{V}_{open}$. Applying the transformation of functors $\tilde{F} \to F$ we obtain the element of $F(\tilde{V}_{\lambda,open})$ we were looking for. This reduces us to the case discussed in the next paragraph.

Proof of surjectivity: the case of an affine W. We have $v \in Z \subset V$ and W is an affine formal algebraic space. Recall that

$$\xi = (Z, u', \hat{x}) \in F(V)$$

We may still replace V by an étale neighbourhood of v. In particular we may and do assume V and V_λ are affine.

Proof of surjectivity: descending Z. We can find a λ and a closed subscheme $Z_\lambda \subset V_\lambda$ such that Z is the base change of Z_λ to V. See Limits, Lemma 01ZM. Warning: we don't know (and in general it won't be true) that Z_λ is a reduced closed subscheme of V_λ. For $\mu \geq \lambda$ denote $Z_\mu \subset V_\mu$ the scheme theoretic inverse image in V_μ. We will use below that $Z = \lim_{\mu \geq \lambda} Z_\mu$ as schemes.

Proof of surjectivity: descending u'. Since U' is locally of finite type over S we may assume after increasing λ that there exists a morphism $u'_\lambda : V_\lambda \setminus Z_\lambda \to U'$ whose restriction to $V \setminus Z$ is u'. See Limits, Proposition 01ZC. For $\mu \geq \lambda$ we will denote u'_μ the restriction of u'_λ to $V_\mu \setminus Z_\mu$.

Proof of surjectivity: descending a witness. Let $V' \to V$, $\hat{x}'$, and x' witness the compatibility between u' and $\hat{x}$. Using the same references as above we may assume (after increasing λ) that there exists a morphism $V'_\lambda \to V_\lambda$ of finite type whose base change to V is $V' \to V$. After increasing λ we may assume $V'_\lambda \to V_\lambda$ is proper (Limits, Lemma 081F). Next, we may assume $V'_\lambda \to V_\lambda$ is an isomorphism over $V_\lambda \setminus Z_\lambda$ (Limits, Lemma 081E). Next, we may assume there is a morphism $x'_\lambda : V'_\lambda \to X'$ whose restriction to V' is x'. Increasing λ again we may assume x'_λ agrees with u'_λ over $V_\lambda \setminus Z_\lambda$. For $\mu \geq \lambda$ we denote V'_μ and x'_μ the base change of V'_λ and the restriction of x'_λ.

Proof of surjectivity: algebra. Write $W = \mathrm{Spf}(B)$, $V = \mathrm{Spec}(A)$, and for $\mu \geq \lambda$ write $V_\mu = \mathrm{Spec}(A_\mu)$. Denote $I_\mu \subset A_\mu$ and $I \subset A$ the ideals cutting out Z_μ and Z. Then $I_\lambda A_\mu = I_\mu$ and $I_\lambda A = I$. The morphism $\hat{x}$ determines and is determined by a continuous ring map

$$(\hat{x})^\sharp : B \longrightarrow A^\wedge$$

where $A^\wedge$ is the I-adic completion of A. To finish the proof we have to show that this map descends to a map into $A^\wedge_\mu$ for some sufficiently large μ where $A^\wedge_\mu$ is the I_μ-adic completion of A_μ. This is a nontrivial fact; Artin writes in his paper [1]: "Since the data (3.5) involve I-adic completions, which do not commute with direct limits, the verification is somewhat delicate. It is an algebraic analogue of a convergence proof in analysis."

Proof of surjectivity: algebra, more rings. Let us denote

$$C_\mu = \Gamma(V'_\mu, \mathcal{O}) \quad \text{and} \quad C = \Gamma(V', \mathcal{O})$$

Observe that $A \to C$ and $A_\mu \to C_\mu$ are finite ring maps as $V' \to V$ and $V'_\mu \to V_\mu$ are proper morphisms, see Cohomology of Spaces, Lemma 08AS. Since $V = \lim V_\mu$ and $V' = \lim V'_\mu$ we have

$$A = \mathrm{colim}\, A_\mu \quad \text{and} \quad C = \mathrm{colim}\, C_\mu$$

by Limits, Lemma 01Z0[4]. For an element $a \in I$, resp. $a \in I_\mu$ the maps $A_a \to C_a$, resp. $(A_\mu)_a \to (C_\mu)_a$ are isomorphisms by flat base change (Cohomology of Spaces, Lemma 073K). Hence the kernel and cokernel of $A \to C$ is supported on $V(I)$ and similarly for $A_\mu \to C_\mu$. We conclude the kernel and cokernel of $A \to C$ are annihilated by a power of I and the kernel and cokernel of $A_\mu \to C_\mu$ are annihilated by a power of I_μ, see Algebra, Lemma 00L6.

Proof of surjectivity: algebra, more ring maps. Denote $Z_n \subset V$ the nth infinitesimal neighbourhood of Z and denote $Z_{\mu,n} \subset V_\mu$ the nth infinitesimal neighbourhood of Z_μ. By the theorem on formal functions (Cohomology of Spaces, Theorem 08AZ) we have

$$C^\wedge = \lim_n H^0(V' \times_V Z_n, \mathcal{O}) \quad \text{and} \quad C_\mu^\wedge = \lim_n H^0(V'_\mu \times_{V_\mu} Z_{\mu,n}, \mathcal{O})$$

where $C^\wedge$ and $C_\mu^\wedge$ are the completion with respect to I and I_μ. Combining the completion of the morphism $x'_\mu : V'_\mu \to X'$ with the morphism $g : X'_{/T'} \to W$ we obtain

$$g \circ x'_{\mu,/Z_\mu} : V'_{\mu,/Z_\mu} = \operatorname{colim} V'_\mu \times_{V_\mu} Z_{\mu,n} \longrightarrow W$$

and hence by the description of the completion $C_\mu^\wedge$ above we obtain a continuous ring homomorphism

$$(g \circ x'_{\mu,/Z_\mu})^\sharp : B \longrightarrow C_\mu^\wedge$$

The fact that $V' \to V$, $\hat{x}'$, x' witnesses the compatibility between u' and $\hat{x}$ implies the commutativity of the following diagram

$$\begin{array}{ccc} C_\mu^\wedge & \longrightarrow & C^\wedge \\ {\scriptstyle (g \circ x'_{\mu,/Z_\mu})^\sharp}\uparrow & & \uparrow \\ B & \xrightarrow{(\hat{x})^\sharp} & A^\wedge \end{array}$$

Proof of surjectivity: more algebra arguments. Recall that the finite A-modules $\operatorname{Ker}(A \to C)$ and $\operatorname{Coker}(A \to C)$ are annihilated by a power of I and similarly the finite A_μ-modules $\operatorname{Ker}(A_\mu \to C_\mu)$ and $\operatorname{Coker}(A_\mu \to C_\mu)$ are annihilated by a power of I_μ. This implies that these modules are equal to their completions. Since I-adic completion on the category of finite A-modules is exact (see Algebra, Section 0BNH) it follows that we have

$$\operatorname{Coker}(A^\wedge \to C^\wedge) = \operatorname{Coker}(A \to C)$$

and similarly for kernels and for the maps $A_\mu \to C_\mu$. Of course we also have

$$\operatorname{Ker}(A \to C) = \operatorname{colim} \operatorname{Ker}(A_\mu \to C_\mu) \quad \text{and} \quad \operatorname{Coker}(A \to C) = \operatorname{colim} \operatorname{Coker}(A_\mu \to C_\mu)$$

Recall that $S = \operatorname{Spec}(R)$ is affine. All of the ring maps above are R-algebra homomorphisms as all of the morphisms are morphisms over S. By Algebraization of Formal Spaces, Lemma 0GC6 we see that B is topologically of finite type over R. Say B is topologically generated by $b_1, \ldots, b_n$. Pick some μ (for example λ) and consider the elements

$$\text{images of } (g \circ x'_{\mu,/Z_\mu})^\sharp(b_1), \ldots, (g \circ x'_{\mu,/Z_\mu})^\sharp(b_n) \text{ in } \operatorname{Coker}(A_\mu \to C_\mu)$$

The image of these elements in $\operatorname{Coker}(\alpha)$ are zero by the commutativity of the square above. Since $\operatorname{Coker}(A \to C) = \operatorname{colim} \operatorname{Coker}(A_\mu \to C_\mu)$ and these cokernels are equal to their completions we see that after increasing μ we may assume these images are all zero. This means that the continuous homomorphism $(g \circ x'_{\mu,/Z_\mu})^\sharp$ has image contained in $\operatorname{Im}(A_\mu \to C_\mu)$. Choose elements $a_{\mu,j} \in (A_\mu)^\wedge$ mapping to $(g \circ x'_{\mu,/Z_\mu})^\sharp(b_1)$ in $(C_\mu)^\wedge$. Then $a_{\mu,j} \in A_\mu^\wedge$ and $(\hat{x})^\sharp(b_j) \in A^\wedge$ map to the same

[4]We don't know that $C_\mu = C_\lambda \otimes_{A_\lambda} A_\mu$ as the various morphisms aren't flat.

element of $C^\wedge$ by the commutativity of the square above. Since $\mathrm{Ker}(A \to C) = \mathrm{colim}\,\mathrm{Ker}(A_\mu \to C_\mu)$ and these kernels are equal to their completions, we may after increasing μ adjust our choices of $a_{\mu,j}$ such that the image of $a_{\mu,j}$ in $A^\wedge$ is equal to $(\hat{x})^\sharp(b_j)$.

Proof of surjectivity: final algebra arguments. Let $\mathfrak{b} \subset B$ be the ideal of topologically nilpotent elements. Let $J \subset R[x_1, \ldots, x_n]$ be the ideal consisting of those $h(x_1, \ldots, x_n)$ such that $h(b_1, \ldots, b_n) \in \mathfrak{b}$. Then we get a continuous surjection of topological R-algebras

$$\Phi : R[x_1, \ldots, x_n]^\wedge \longrightarrow B, \quad x_j \longmapsto b_j$$

where the completion on the left hand side is with respect to J. Since $R[x_1, \ldots, x_n]$ is Noetherian we can choose generators $h_1, \ldots, h_m$ for J. By the commutativity of the square above we see that $h_j(a_{\mu,1}, \ldots, a_{\mu,n})$ is an element of $A_\mu^\wedge$ whose image in $A^\wedge$ is contained in $IA^\wedge$. Namely, the ring map $(\hat{x})^\sharp$ is continuous and $IA^\wedge$ is the ideal of topological nilpotent elements of $A^\wedge$ because $A^\wedge/IA^\wedge = A/I$ is reduced. (See Algebra, Section 0BNH for results on completion in Noetherian rings.) Since $A/I = \mathrm{colim}\, A_\mu/I_\mu$ we conclude that after increasing μ we may assume $h_j(a_{\mu,1}, \ldots, a_{\mu,n})$ is in $I_\mu A_\mu^\wedge$. In particular the elements $h_j(a_{\mu,1}, \ldots, a_{\mu,n})$ of $A_\mu^\wedge$ are topologically nilpotent in $A_\mu^\wedge$. Thus we obtain a continuous R-algebra homomorphism

$$\Psi : R[x_1, \ldots, x_n]^\wedge \longrightarrow A_\mu^\wedge, \quad x_j \longmapsto a_{\mu,j}$$

In order to conclude what we want, we need to see if $\mathrm{Ker}(\Phi)$ is annihilated by Ψ. This may not be true, but we can achieve this after increasing μ. Indeed, since $R[x_1, \ldots, x_n]^\wedge$ is Noetherian, we can choose generators $g_1, \ldots, g_l$ of the ideal $\mathrm{Ker}(\Phi)$. Then we see that

$$\Psi(g_1), \ldots, \Psi(g_l) \in \mathrm{Ker}(A_\mu^\wedge \to C_\mu^\wedge) = \mathrm{Ker}(A_\mu \to C_\mu)$$

map to zero in $\mathrm{Ker}(A \to C) = \mathrm{colim}\,\mathrm{Ker}(A_\mu \to C_\mu)$. Hence increasing μ as before we get the desired result.

Proof of surjectivity: mopping up. The continuous ring homomorphism $B \to (A_\mu)^\wedge$ constructed above determines a morphism $\hat{x}_\mu : V_{\mu,/Z_\mu} \to W$. The compatibility of $\hat{x}_\mu$ and u'_μ follows from the fact that the ring map $B \to (A_\mu)^\wedge$ is by construction compatible with the ring map $A_\mu \to C_\mu$. In fact, the compatibility will be witnessed by the proper morphism $V'_\mu \to V_\mu$ and the morphisms x'_μ and $\hat{x}'_\mu = x'_{\mu,/Z_\mu}$ we used in the construction. This finishes the proof. □

References

[1] Michael Artin. Algebraization of formal moduli: II – existence of modifications. *Annals of Mathematics*, 91:88–135, 1970.